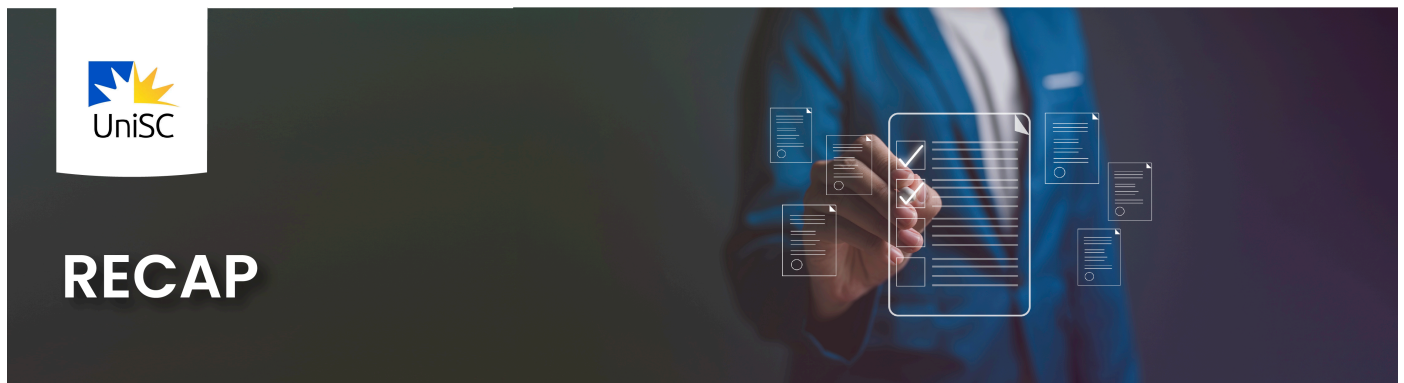


Module 1 Summary



What You Learned in Module 1

1. Financial Analytics and the Role of Time

You learned that:

- Financial analytics is about using data to inform financial decisions.
- Time is central because financial variables evolve, compound, and respond to events.
- Financial data is naturally structured as time series.

You now understand that analysing financial data requires thinking about *how* values change over time, not just what they are at a single point.

This directly supports:

You can now explain why time-series analysis is essential in finance.

2. Getting Started with R and RStudio

You learned to:

- Install R and RStudio.
- Navigate the four RStudio panes (Source, Console, Environment, Files/Plots/Packages/Help).
- Write and run scripts instead of typing commands manually.

-
- Store results as objects and locate them in the Environment pane.
 - Install packages once and load them each session.

This supports:

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

You can now use R as a practical tool for financial data analysis.

3. Accessing and Visualising Financial Time Series

You learned to:

- Download stock price data using `tq_get()`.
- Understand what the returned tibble contains (dates, prices, volumes, etc.).
- Visualise price series using `ggplot()`.
- Interpret basic trends, patterns, and movements in price charts.

You now understand what price charts show — and what they do *not* show (e.g., returns, risk, or growth rates).

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You have begun applying time-series reasoning to real financial data.

4. Computing Returns and Growth Rates

You learned to:

- Compute monthly and annual returns using `tq_transmute()`.
- Interpret what “return frequency” means.
- Compute CAGR using start value, end value, and elapsed time.
- Compare growth across assets (e.g., Microsoft, Apple, housing index).

You now understand that returns summarise *changes* in value, while CAGR summarises long-term growth.

This supports:

CLO2 and CLO3, as you applied quantitative techniques and interpreted them in financial contexts.

5. APIs and Accessing Economic Data

You learned:

- What an API is and why it matters for reliable data access.
- How to obtain and set a FRED API key.
- How to download economic time series (e.g., Australian house price index).
- How to compute and compare CAGR across different asset classes.

You now understand how to access high-quality data programmatically and why this reduces errors.

This supports:

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights.

Practice Tasks

Complete the following tasks in RStudio using a script. You can use the R scripts provided throughout this module.

Practice Task 1: Setup and packages

1. Install the packages used in this module in one line of code.
2. Load the packages for your session.
3. In one or two sentences, explain the difference between installing a package and loading a package.

Practice Task 2: Download and inspect stock data

1. Use `tq_get()` to download stock price data for any two stocks that interest you
2. Print the first few rows of each dataset.
3. Write down the column name you used for price in your plots and return calculations. Explain why adjusted prices are typically preferred.

Practice Task 3: Visualise prices

1. Create a line chart of the stock prices
2. Create one chart that shows both stocks together, using colour to distinguish the series.
3. Write 3 to 5 sentences comparing the two price series. Focus on what you can observe from the chart, and be clear about what you cannot infer from price alone.

Practice Task 4: Calculate returns at different frequencies

1. Using `tq_transmute()`, compute:
 - Monthly returns
 - Annual returns
2. For each stock, compute:
 - The mean monthly return

- The mean annual return

Practice Task 5: Calculate CAGR for both stocks

1. Compute CAGR for the stocks from the first available date in your dataset to the last available date.
2. In one paragraph, explain what CAGR measures and what information it hides

Practice Task 6: FRED data and house price index comparison

1. In 2 to 4 sentences, explain what an API is and what an API key does.
2. Obtain a FRED API key and set it using `fredr_set_key()`.
3. Import the data for the S&P 500 (SP500)
4. Compute the CAGR of the index over the available sample.

Extend your knowledge:

1. How would you invest across these stocks and the SP500.
2. Write a short interpretation (5 to 8 sentences). Your interpretation should:
 - Identify which asset had the highest CAGR and which had the lowest
 - Comment on at least one limitation of comparing these series using only CAGR

Reflections

Write short responses to the following prompts. Aim for clear, concrete answers, not technical detail.

1. What is one task you currently do in Excel that could become more reliable or repeatable if you did it in R instead? What would be the benefit?
2. Pick one financial concept from this module (returns, CAGR, price index, or time series). How does the fact that it unfolds over time affect how you interpret it?
3. When you plotted Microsoft and Apple prices, what did you notice about the series? What does a price chart tell you, and what does it not tell you?
4. Why might downloading data via packages and APIs be preferable to manually exporting and copying data? Where could errors or inconsistencies enter if you rely on manual steps?

Workshop

[Tasks \(https://learn.usc.edu.au/courses/35359/files/3282923?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3282923?wrap=1)